

Homework 11 Tuesday

Sunday, 17 March 2024 5:28 pm

- Express the product $A\mathbf{x}$ as a linear combination of the column vectors of A .
- 2.

$$(a) \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$(b) \begin{bmatrix} 4 & 0 & -1 \\ 3 & 6 & 2 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix}$$

Question 02:

Part (a)

(a)

$$\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + 3 \cdot 2 \\ (-1) \cdot 1 + 4 \cdot 2 \end{bmatrix} = 1 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Part (b)

(b)

$$\begin{bmatrix} 4 & 0 & -1 \\ 3 & 6 & 2 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \cdot (-2) + 0 \cdot 3 + (-1) \cdot 5 \\ 3 \cdot (-2) + 6 \cdot 3 + 2 \cdot 5 \\ 0 \cdot (-2) + (-1) \cdot 3 + 4 \cdot 5 \end{bmatrix} = (-2) \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 6 \\ -1 \end{bmatrix} + 5 \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}$$

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- Find a basis for the nullspace of A .
- 6.

$$(a) A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix}$$

Part (a)

Step 1

1 of 7

(a)

Reducing matrix

$$\begin{bmatrix} 1 & -1 & 3 \\ 5 & -4 & -4 \\ 7 & -6 & 2 \end{bmatrix} \xrightarrow{(1)} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 1 & -19 \end{bmatrix} \xrightarrow{(2)} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{(3)} \begin{bmatrix} 1 & 0 & -16 \\ 0 & 1 & -19 \\ 0 & 0 & 0 \end{bmatrix}$$

Explanations:

(1) Multiply row 1 by -5 and add to row 2. Multiply row 1 by -7 and add to row 3.

(2) Subtract row 2 from row 3.

(3) Add row 2 to row 1.

So if

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

is solution of the system $A\mathbf{x} = \mathbf{0}$ then

$$x_1 - 16x_3 = 0$$

$$x_2 - 19x_3 = 0$$

If we set $x_3 = t$ then $x_1 = 16t$, $x_2 = 19t$ and

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = t \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix}$$

Therefore

$$\left\{ \begin{bmatrix} 16 \\ 19 \\ 1 \end{bmatrix} \right\}$$

is basis for the null space of A .

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$$(c) \quad A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$$

Reducing matrix

$$\begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix} \xrightarrow{(1)} \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & -7 & -7 & -4 \\ 0 & 7 & 7 & 4 \end{bmatrix} \xrightarrow{(2)} \begin{bmatrix} 1 & 4 & 5 & 2 \\ 0 & -7 & -7 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{(3)} \begin{bmatrix} 2 & 1 & 3 & 0 \\ 0 & -7 & -7 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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$A\mathbf{x} = \mathbf{0}$ then

$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 0 \\ -7x_2 - 7x_3 - 4x_4 &= 0 \end{aligned}$$

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If we set $x_2 = t$, $x_3 = s$ then $x_1 = -\frac{1}{2}t - \frac{3}{2}s$, $x_4 = -\frac{7}{4}t - \frac{7}{4}s$ and

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}t - \frac{3}{2}s \\ t \\ s \\ -\frac{7}{4}t - \frac{7}{4}s \end{bmatrix} = t \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ -\frac{7}{4} \end{bmatrix} + s \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 1 \\ -\frac{7}{4} \end{bmatrix}$$

Therefore

$$\left\{ \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \\ -\frac{7}{4} \end{bmatrix}, \begin{bmatrix} -\frac{3}{2} \\ 0 \\ 1 \\ -\frac{7}{4} \end{bmatrix} \right\}$$

is basis for the null space of A .

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13. Prove that the row vectors of an $n \times n$ invertible matrix A form a basis for $\mathbb{R}^n$.

- 13.** Let A be an $n \times n$ invertible matrix. Since A^T is also invertible, it is row equivalent to I_n . It is clear that the column vectors of I_n are linearly independent. Hence, by virtue of Theorem 5.5.5, the column vectors of A^T , which are just the row vectors of A , are also linearly independent. Therefore the rows of A form a set of n linearly independent vectors in $\mathbb{R}^n$, and consequently form a basis for $\mathbb{R}^n$.

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